

Deep Learning Surrogate Model for Rapid Aerodynamic Coefficient Prediction

Sanket Undre (Matr. Number: 12347493)
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1 Abstract

This report details the development of a Deep Learning Surrogate Model designed to predict aerodynamic coefficients (C_l and C_d) from airfoil geometry. Using a Multi-Layer Perceptron (MLP) trained on the DeepLearWing dataset, the model acts as a high-speed alternative to Computational Fluid Dynamics (CFD). A key contribution of this work is a robust data re-sampling pipeline and a dynamic training strategy that significantly improved Drag prediction accuracy ($R^2 = 0.91$). Final validation on unseen real-world airfoils (NACA 2414, NACA 6409) demonstrates the model's capability to generalize well beyond the training distribution.

2 Problem Statement and Solution

2.1 What is the problem and why is it significant?

In the early stages of aerodynamic design, engineers must evaluate thousands of potential airfoil shapes to optimize performance. The core problem is the high computational cost of these evaluations. Traditional methods rely on CFD solvers (like XFOIL or OpenFOAM) which solve the Navier-Stokes equations. While accurate, these simulations are computationally expensive and slow, often taking minutes per simulation. This creates a significant bottleneck in optimization loops, limiting the number of designs an engineer can explore in a given timeframe.

2.2 What is the solution and why Deep Learning?

The solution developed is a deep learning pipeline that maps geometric coordinates and flow conditions (Reynolds number, Angle of Attack) directly to aerodynamic forces.

Why is this a solution? Deep Learning is particularly well-suited for this task because it acts as a universal function approximator. Unlike CFD, which must solve complex differential equations from scratch for every new input, the neural network learns the non-linear relationship $f(\text{geometry, physics}) \rightarrow (C_l, C_d)$ once.

- **Inference Speed:** Once trained, the model reduces the time-to-result from minutes to milliseconds.
- **Differentiability:** Neural networks are differentiable, which allows them to be easily integrated into gradient-based design optimization loops in the future.

3 Methodology

3.1 Data Preprocessing and Resampling

A critical challenge identified early in the project was inconsistent input data; raw airfoil files contained varying numbers of coordinate points. To address this, a custom resampling algorithm was implemented (visualized in Figure 1).

- **Geometric Normalization:** Every input airfoil is resampled to exactly 50 discrete coordinate points (100 features) using linear interpolation based on cumulative chord distance.
- **Physics Scaling:** The Reynolds number (Re), spanning orders of magnitude ($50k - 1M$), was log-transformed ($\log_{10}(Re)$) to prevent gradient instability.

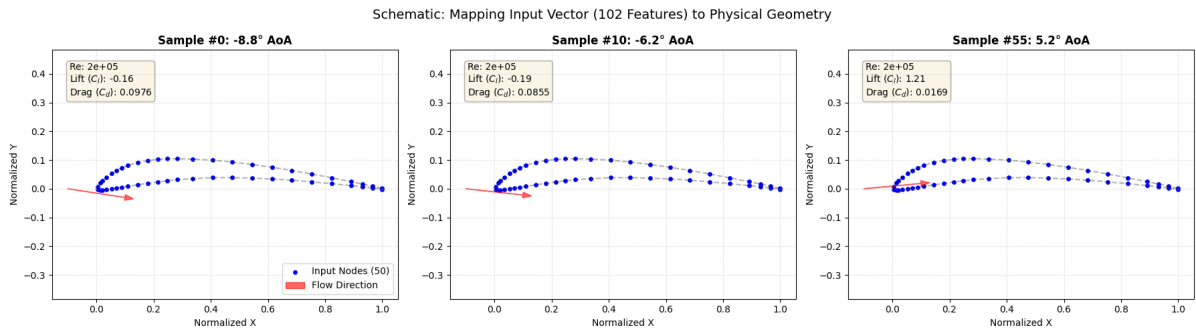


Figure 1: Data Processing Pipeline: Raw geometries are normalized to a fixed 102-dimensional input vector (100 geometric nodes + AoA + Reynolds).

3.2 Network Architecture

The model is a Multi-Layer Perceptron (MLP) with a "funnel" architecture designed to compress geometric features into scalar outputs.

- **Structure:** Input (102) $\rightarrow$ FC(256) $\rightarrow$ FC(128) $\rightarrow$ FC(64) $\rightarrow$ Output(2).
- **Components:** Rectified Linear Units (ReLU) introduce non-linearity, and Batch Normalization is applied after the first hidden layer to stabilize weight updates.

4 Experiments and Results

4.1 Training Dynamics and Optimization

Predicting Drag (C_d) proved difficult due to its small magnitude and complex non-linear behavior near stall. Initial experiments yielded a poor R^2 of 0.79 for Drag. **Key Insight:** Implementing a Learning Rate Scheduler (`ReduceLROnPlateau`) was a critical step. As shown in Figure 2, the scheduler reduced the learning rate when validation loss plateaued (around epoch 5 and 12), allowing the optimizer to settle into a deeper minimum and capture the subtle physics of Drag.

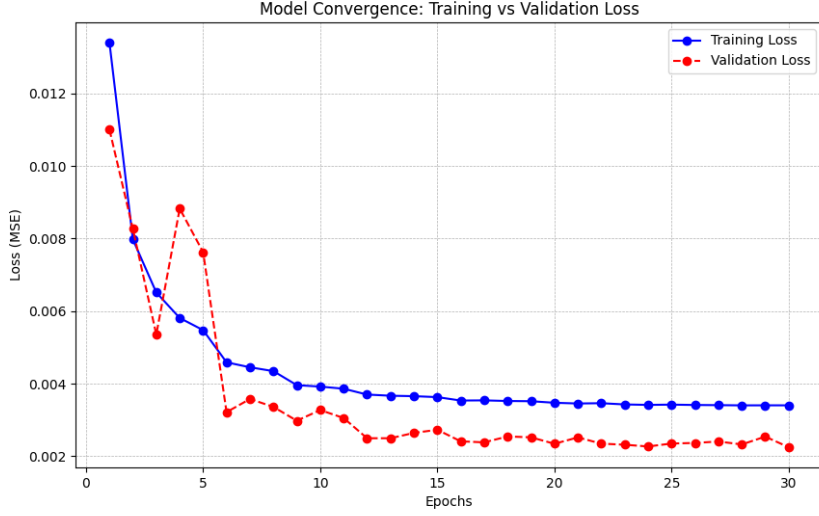


Figure 2: Training Convergence: The "steps" in the loss curve indicate where the scheduler reduced the learning rate, leading to significant error reduction.

4.2 Test Set Evaluation

The model's performance was evaluated on a held-out test set of 5,000 samples. Two metrics were calculated: the Coefficient of Determination (R^2) and the Mean Absolute Error (MAE). Table 1 presents a comparison between the initial experimental run (without scheduling) and the final optimized pipeline.

Table 1: Comparison of Model Performance

Target	Initial Experiment (R^2)	Final Model (R^2)	Final MAE
Lift (C_l)	0.98	0.99	0.0340
Drag (C_d)	0.79	0.91	0.0068

The improvement in Drag prediction is substantial. Figure 3 provides a regression plot of predicted versus actual values.

- **Lift Analysis:** The predictions for Lift (left plot) are nearly perfectly aligned with the diagonal $y = x$ line. This confirms that the MLP has effectively learned the linear physics of lift generation.
- **Drag Analysis:** The predictions for Drag (right plot) show a tight clustering around the diagonal. Crucially, the model accurately predicts values in the high-drag region ($C_d > 0.10$), which typically corresponds to stall conditions. This area was poorly predicted in the initial experiments, validating the effectiveness of the optimization strategy.

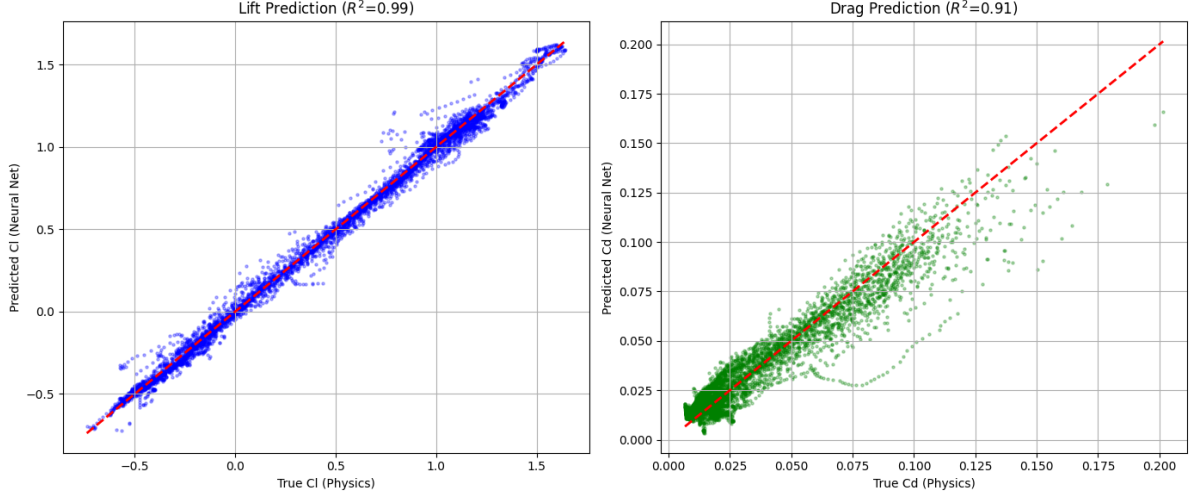


Figure 3: Final regression results on the test set. The tight correlation, particularly for Drag, demonstrates the model’s ability to capture complex aerodynamic phenomena.

4.3 Generalization to Unseen Airfoils

To verify that the model is not simply memorizing the training set, a "Generalization Test" was conducted. Two standard airfoils (NACA 2414 and NACA 6409) were downloaded from *airfoiltools.com*. These geometries were **not** in the training set. The model’s predictions were compared against XFOIL ground truth data.

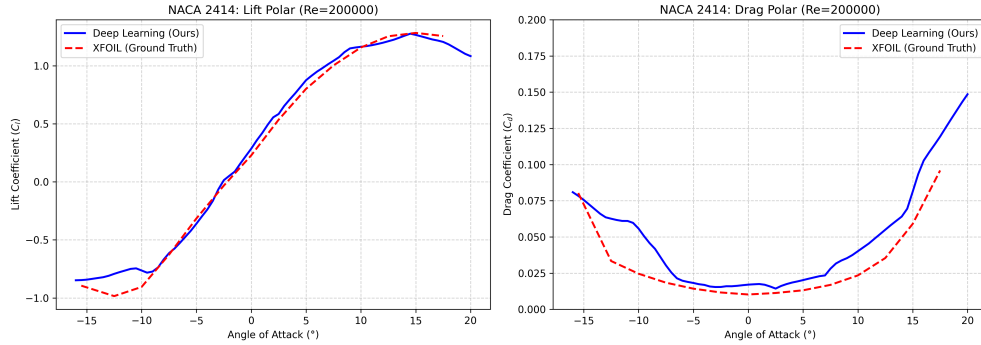


Figure 4: Validation on NACA 2414 (Unseen Data). The model captures the linear lift slope and the drag bucket accurately.

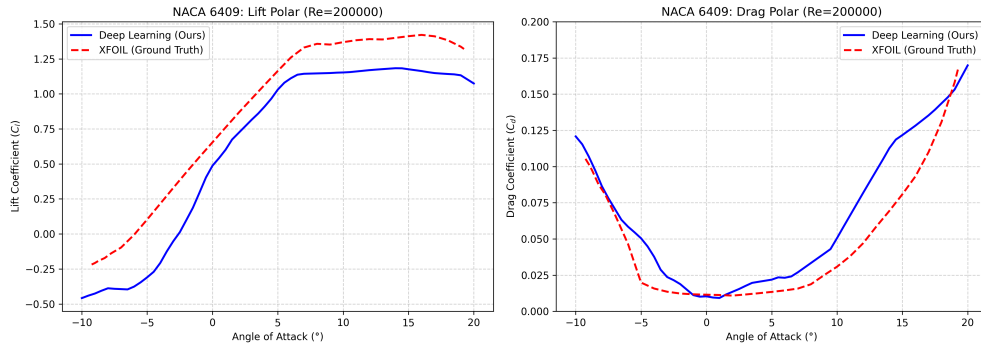


Figure 5: Validation on NACA 6409 (Unseen Data). The deep learning predictions closely track the XFOIL ground truth across the entire Angle of Attack range.

5 Discussion, Insights, and Reflection

5.1 Key Insights

Throughout the project, several technical insights emerged:

- **Data Quality & Model Complexity:** Standardizing the input geometry via interpolation provided a larger performance boost than simply adding layers. The model could not learn effectively when input vectors represented different topological densities.
- **Optimization Strategy:** The use of a dynamic Learning Rate Scheduler was the single most effective "hack" for improving Drag prediction. The non-convex loss landscape for Drag required a finer learning rate to converge than Lift.

5.2 Reflection: What can be done differently?

If this project is repeated, then the exploration using the **1D Convolutional Neural Networks (CNNs)** would be a best fit. Since airfoil coordinates have spatial relationships (neighbors matter), a CNN might capture local curvature features more efficiently than the global fully-connected layers of an MLP. Additionally, another method would be to attempt "Inverse Design." Currently, the model predicts performance (C_l, C_d) from geometry. A more powerful tool would be the reverse: predicting the optimal airfoil shape from desired aerodynamic properties, potentially using Generative Adversarial Networks (GANs).

5.3 Time Investment

I spent approximately **54 hours** on this project. This aligns closely with my initial estimate of 60-70 hours. However, the distribution of time was unexpected; the majority was not spent on model design, but on the "Hacking" phase—specifically debugging data inconsistencies and tuning the scheduler to fix the Drag prediction. I underestimated the complexity of preprocessing the raw geometry files.

6 Conclusion

This project explored the feasibility of using Deep Learning as a surrogate for aerodynamic simulations. By addressing the challenges of data irregularity and non-linear regression, the final pipeline produced a model that closely approximates physics-based solvers. The results on the test set ($R^2 > 0.9$) and the validation on unseen NACA airfoils suggest that the MLP architecture has effectively learned the underlying aerodynamic principles. The current implementation serves as a functional proof-of-concept for real-time aerodynamic analysis, with clear pathways for future improvement in inverse design.